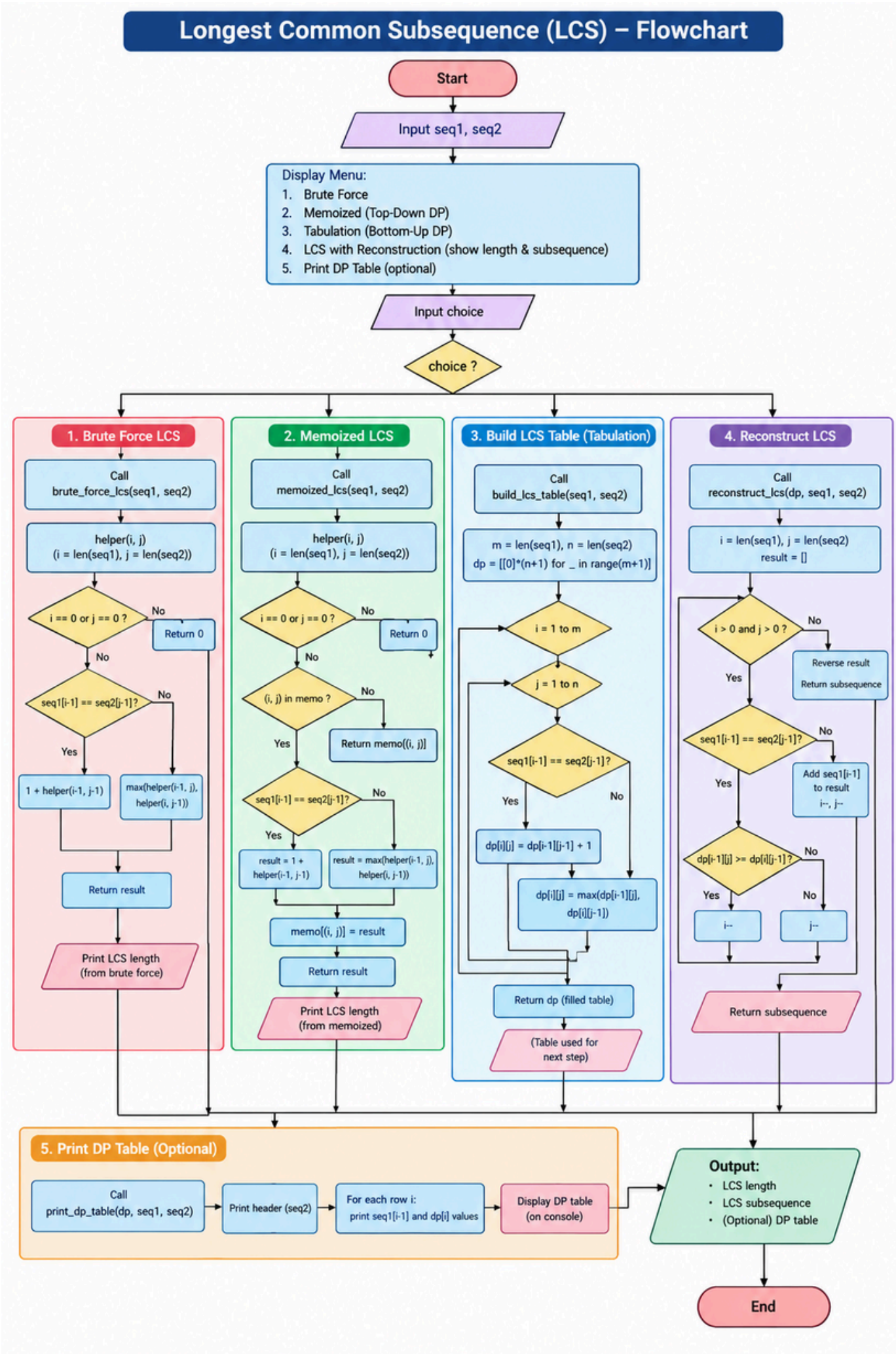
# Algorithm: Longest Common Subsequence (LCS)

**Dynamic Programming (Tabulation), with Brute-Force and Memoized Versions**

1. Start the program.
2. Take two sequences as input.
3. Implement a brute-force recursive method for comparison.
4. Implement a top-down dynamic programming method using memoization.
5. Create a bottom-up DP table with rows for the first sequence and columns for the second sequence.
6. Initialize the DP table with zero values.
7. Compare each character or element of the two sequences.
8. If the elements match, set dp[i][j] = dp[i-1][j-1] + 1.
9. If the elements do not match, set dp[i][j] = max(dp[i-1][j], dp[i][j-1]).
10. Continue until the complete DP table is filled.
11. Take dp[m][n], the bottom-right cell, as the LCS length.
12. Backtrack from the bottom-right corner to reconstruct the actual LCS.
13. If the elements match, add the element to the result and move diagonally.
14. Otherwise, move up or left according to the larger DP value.
15. Reverse the reconstructed result.
16. Display the DP table, LCS length, and LCS subsequence.
17. Cross-check the result using the memoized and brute-force approaches.
18. Demonstrate that LCS also works on lists.
19. End the program.


Longest Common Subsequence (LCS) – Flowchart
Start
Input seq1, seq2
Display Menu:
1. Brute Force
2. Memoized (Top-Down DP)
3. Tabulation (Bottom-Up DP)
4. LCS with Reconstruction (show length & subsequence)
5. Print DP Table (optional)
Input choice
choice ?
1. Brute Force LCS
Call brute_force_lcs(seq1, seq2)
helper(i, j) (i = len(seq1), j = len(seq2))
i == 0 or j == 0 ?
No
Return 0
No
seq1[i-1] == seq2[j-1]?
No
Yes
1 + helper(i-1, j-1)
max(helper(i-1, j), helper(i, j-1))
Return result
Print LCS length (from brute force)
2. Memoized LCS
Call memoized_lcs(seq1, seq2)
helper(i, j) (i = len(seq1), j = len(seq2))
i == 0 or j == 0 ?
No
Return 0
No
(i, j) in memo ?
No
Yes
Return memo[(i, j)]
seq1[i-1] == seq2[j-1]?
No
Yes
result = 1 + helper(i-1, j-1)
result = max(helper(i-1, j), helper(i, j-1))
memo[(i, j)] = result
Return result
Print LCS length (from memoized)
3. Build LCS Table (Tabulation)
Call build_lcs_table(seq1, seq2)
m = len(seq1), n = len(seq2)
dp = [[0]*(n+1) for _ in range(m+1)]
i = 1 to m
j = 1 to n
seq1[i-1] == seq2[j-1]?
Yes
No
dp[i][j] = dp[i-1][j-1] + 1
dp[i][j] = max(dp[i-1][j], dp[i][j-1])
Return dp (filled table)
(Table used for next step)
4. Reconstruct LCS
Call reconstruct_lcs(dp, seq1, seq2)
i = len(seq1), j = len(seq2)
result = []
i > 0 and j > 0 ?
No
Reverse result
Return subsequence
Yes
seq1[i-1] == seq2[j-1]?
No
Yes
Add seq1[i-1] to result
i--, j--
dp[i-1][j] >= dp[i][j-1] ?
No
Yes
No
i--
j--
Return subsequence
5. Print DP Table (Optional)
Call print_dp_table(dp, seq1, seq2)
Print header (seq2)
For each row i: print seq1[i-1] and dp[i] values
Display DP table (on console)
Output:
• LCS length
• LCS subsequence
• (Optional) DP table
End

# Longest Common Subsequence — Python Source Code

```
# LONGEST COMMON SUBSEQUENCE (LCS)

def brute_force_lcs(seq1, seq2):
    def helper(i, j):
        if i == 0 or j == 0:
            return 0
        if seq1[i - 1] == seq2[j - 1]:
            return 1 + helper(i - 1, j - 1)
        return max(helper(i - 1, j), helper(i, j - 1))
    return helper(len(seq1), len(seq2))


def memoized_lcs(seq1, seq2):
    memo = {}
    def helper(i, j):
        if i == 0 or j == 0:
            return 0
        if (i, j) in memo:
            return memo[(i, j)]
        if seq1[i - 1] == seq2[j - 1]:
            result = 1 + helper(i - 1, j - 1)
        else:
            result = max(helper(i - 1, j), helper(i, j - 1))
        memo[(i, j)] = result
        return result
    return helper(len(seq1), len(seq2))


def build_lcs_table(seq1, seq2):
    m, n = len(seq1), len(seq2)
    dp = [[0] * (n + 1) for _ in range(m + 1)]

    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if seq1[i - 1] == seq2[j - 1]:
                dp[i][j] = dp[i - 1][j - 1] + 1
            else:
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
    return dp


def reconstruct_lcs(dp, seq1, seq2):
    i, j = len(seq1), len(seq2)
    result = []

    while i > 0 and j > 0:
        if seq1[i - 1] == seq2[j - 1]:
        result.append(seq1[i - 1]) i -
        = 1
        j -= 1
        elif dp[i - 1][j] >= dp[i][j - 1]:
            i -= 1
        else:
            j -= 1

    result.reverse()
    return "".join(result) if isinstance(seq1, str) else result


def lcs(seq1, seq2):
    dp = build_lcs_table(seq1, seq2)
    length = dp[len(seq1)][len(seq2)]
    subsequence = reconstruct_lcs(dp, seq1, seq2)
    return length, subsequence, dp


def print_dp_table(dp, seq1, seq2):
    header = "        " + "  ".join(f"{c:>2}" for c in seq2)
    print(header)
    for i, row in enumerate(dp):
        label = seq1[i - 1] if i > 0 else " "
        print(f"  {label:>2} | " + " ".join(f"{v:>2}" for v in row))


if __name__ == "__main__":
    seq1 = "ABCBDAB"
    seq2 = "BDCABA"

    print(f"Sequence 1: {seq1}")
    print(f"Sequence 2: {seq2}\n")

    length, subsequence, dp = lcs(seq1, seq2)

    print("DP Table:")
    print_dp_table(dp, seq1, seq2)
```

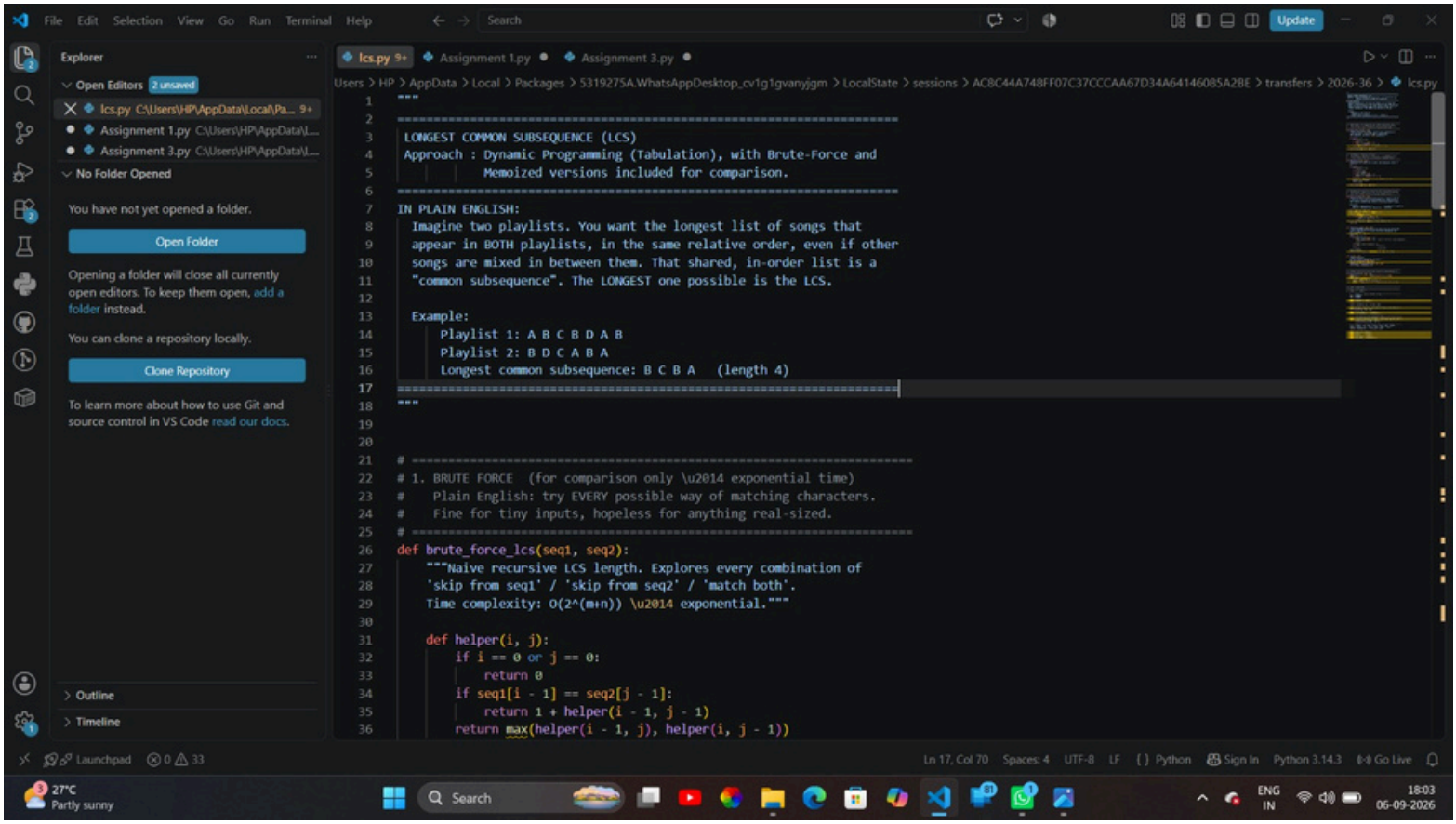
```
print(f"\nLCS length      : {length}")
print(f"LCS subsequence : {subsequence}")

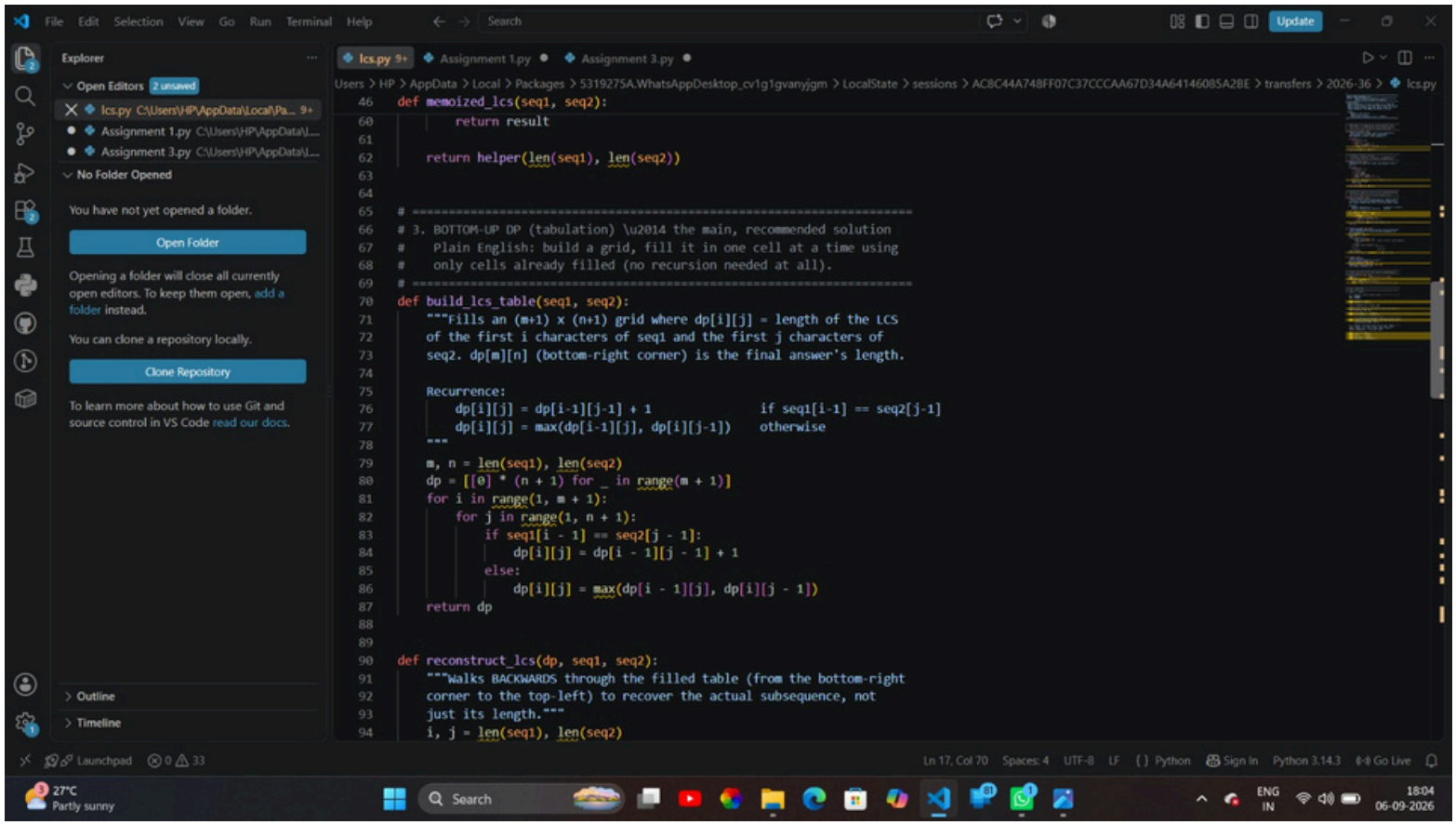
print("\n[check] memoized_lcs length :",
      memoized_lcs(seq1, seq2))
print("[check] brute_force_lcs length (ABC vs AC) :",
      brute_force_lcs("ABC", "AC"))

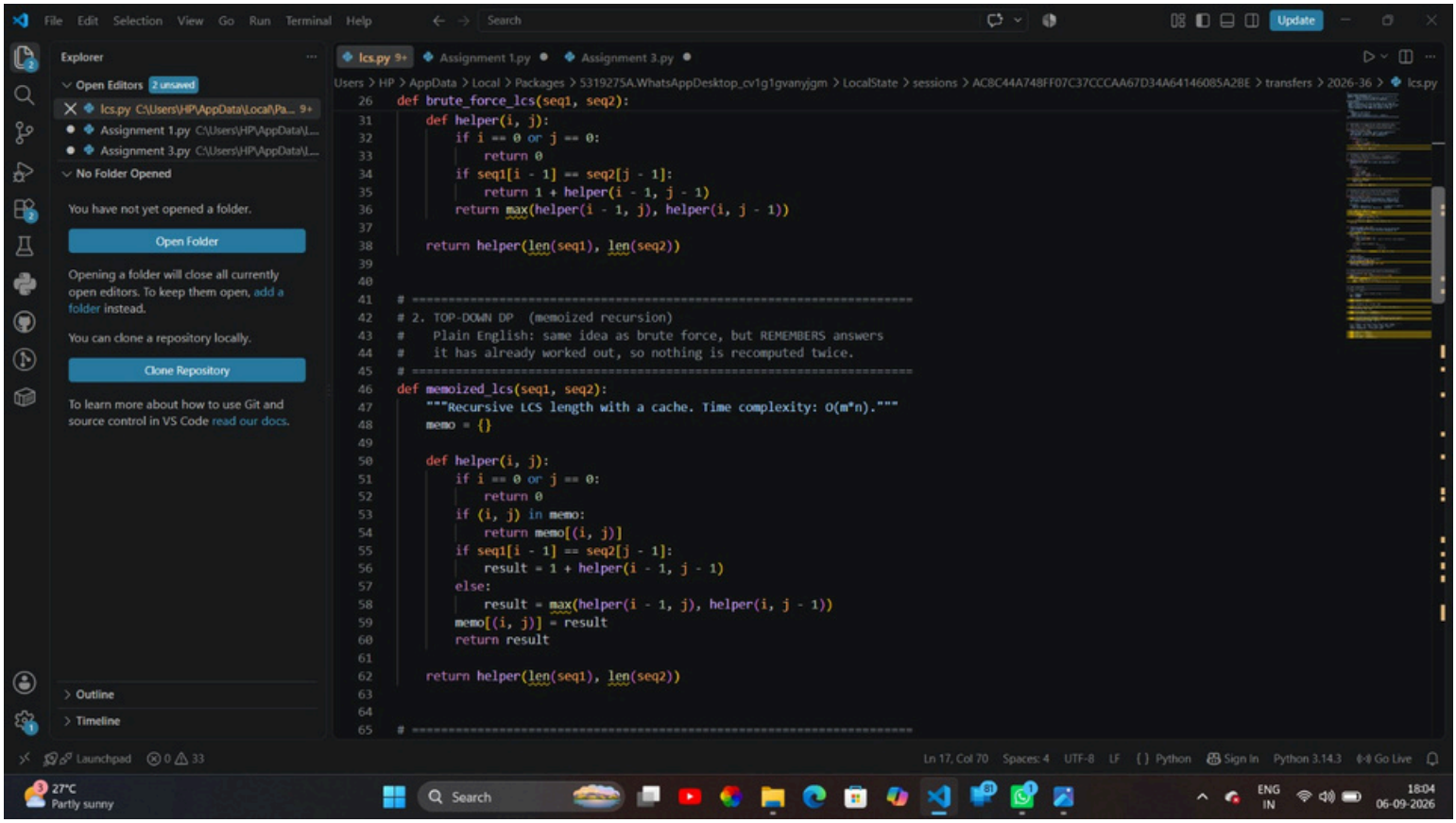
list1 = ["walk", "to", "the", "store", "and", "buy", "milk"]
list2 = ["drive", "to", "store", "buy", "some", "milk"]

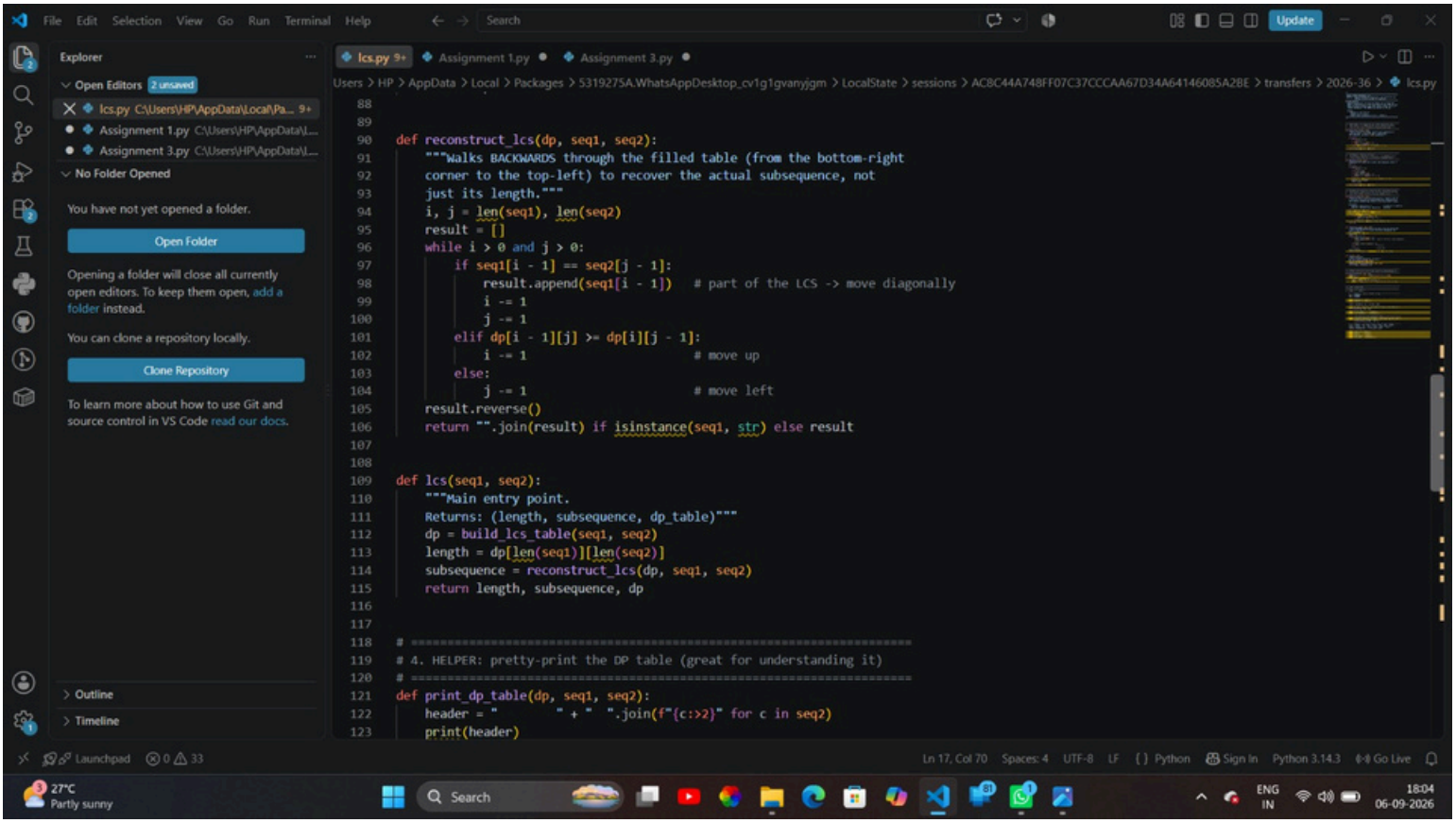
length2, subsequence2, _ = lcs(list1, list2)

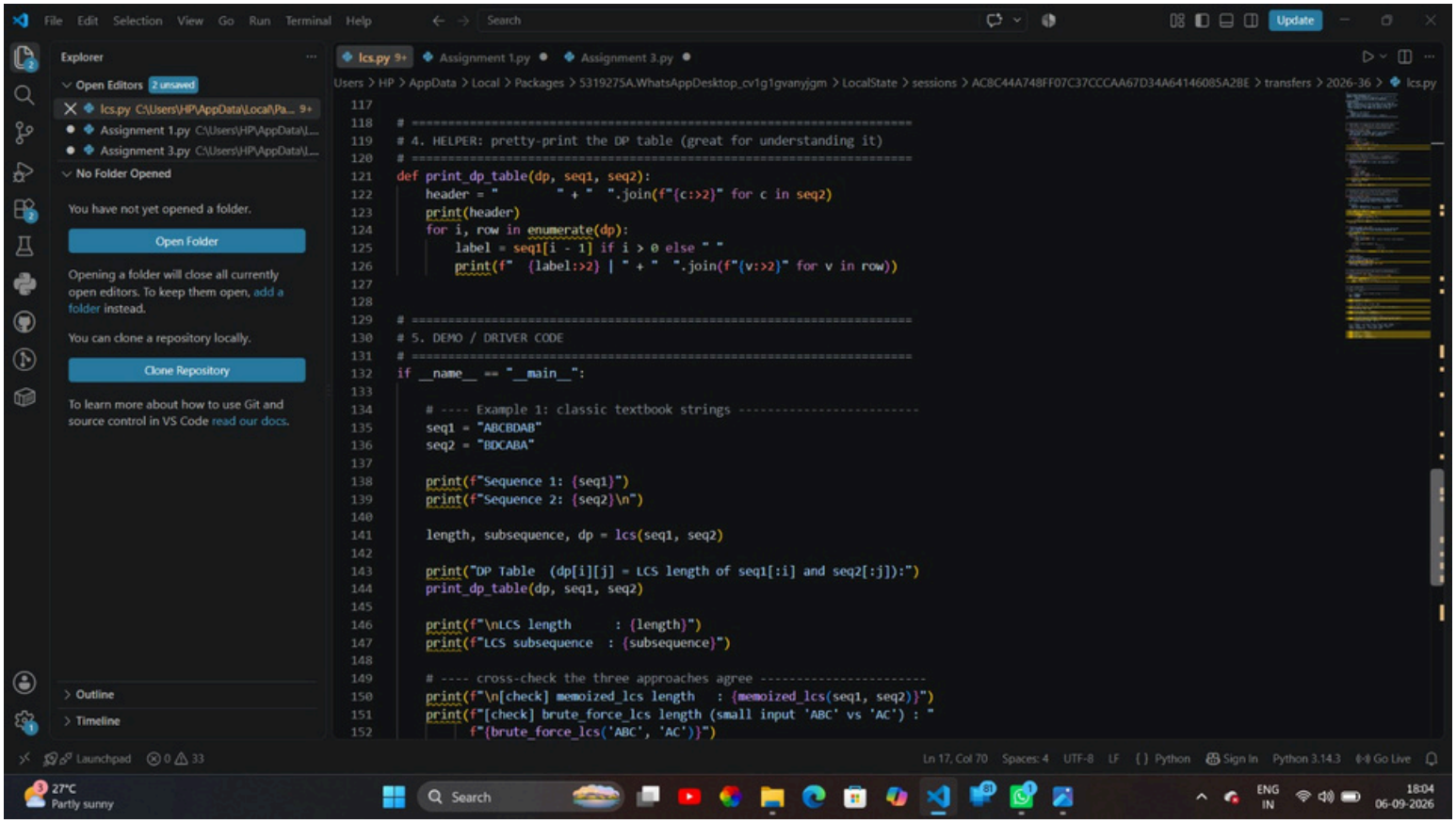
print("\nWord-list example:")
print(f"List  1    : {list1}")
print(f"List 2     : {list2}")
print(f"LCS length : {length2}")
print(f"LCS words  : {subsequence2}")
```

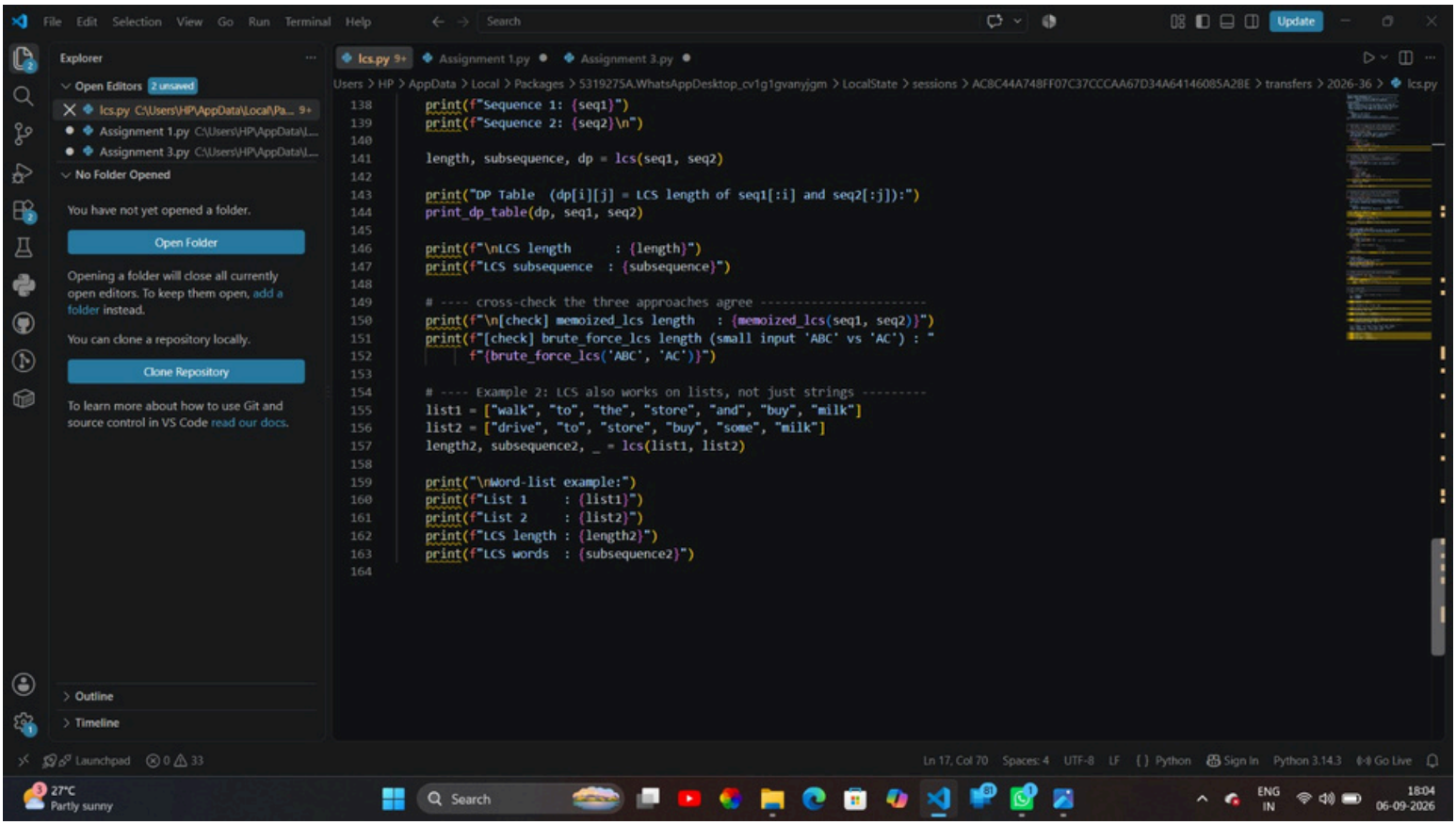

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"""
LONGEST COMMON SUBSEQUENCE (LCS)
Approach : Dynamic Programming (Tabulation), with Brute-Force and
Memoized versions included for comparison.
IN PLAIN ENGLISH:
Imagine two playlists. You want the longest list of songs that
appear in BOTH playlists, in the same relative order, even if other
songs are mixed in between them. That shared, in-order list is a
"common subsequence". The LONGEST one possible is the LCS.
Example:
Playlist 1: A B C B D A B
Playlist 2: B D C A B A
Longest common subsequence: B C B A (length 4)
"""
# 1. BRUTE FORCE (for comparison only \u2014 exponential time)
# Plain English: try EVERY possible way of matching characters.
# Fine for tiny inputs, hopeless for anything real-sized.
def brute_force_lcs(seq1, seq2):
"""Naive recursive LCS length. Explores every combination of
'skip from seq1' / 'skip from seq2' / 'match both'.
Time complexity: O(2^(m+n)) \u2014 exponential."""
def helper(i, j):
if i == 0 or j == 0:
return 0
if seq1[i - 1] == seq2[j - 1]:
return 1 + helper(i - 1, j - 1)
return max(helper(i - 1, j), helper(i, j - 1))
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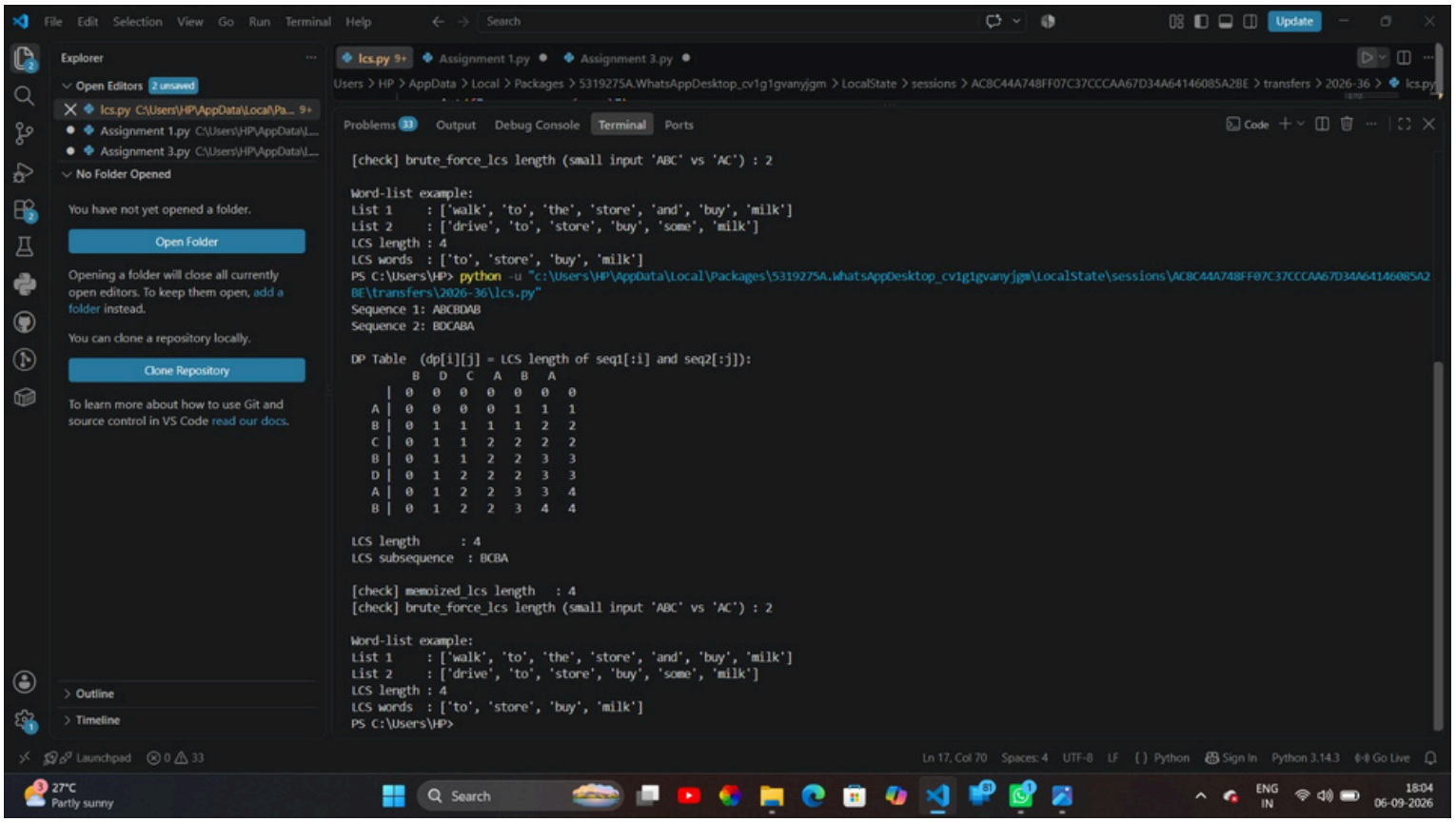

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def memoized_lcs(seq1, seq2):
return result
return helper(len(seq1), len(seq2))
# 3. BOTTOM-UP DP (tabulation) \u2014 the main, recommended solution
# Plain English: build a grid, fill it in one cell at a time using
# only cells already filled (no recursion needed at all).
def build_lcs_table(seq1, seq2):
"""Fills an (m+1) x (n+1) grid where dp[i][j] = length of the LCS
of the first i characters of seq1 and the first j characters of
seq2. dp[m][n] (bottom-right corner) is the final answer's length.
Recurrence:
dp[i][j] = dp[i-1][j-1] + 1 if seq1[i-1] == seq2[j-1]
dp[i][j] = max(dp[i-1][j], dp[i][j-1]) otherwise
"""
m, n = len(seq1), len(seq2)
dp = [[0] * (n + 1) for _ in range(m + 1)]
for i in range(1, m + 1):
for j in range(1, n + 1):
if seq1[i - 1] == seq2[j - 1]:
dp[i][j] = dp[i - 1][j - 1] + 1
else:
dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
return dp
def reconstruct_lcs(dp, seq1, seq2):
"""Walks BACKWARDS through the filled table (from the bottom-right
corner to the top-left) to recover the actual subsequence, not
just its length."""
i, j = len(seq1), len(seq2)
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def brute_force_lcs(seq1, seq2):
def helper(i, j):
if i == 0 or j == 0:
return 0
if seq1[i - 1] == seq2[j - 1]:
return 1 + helper(i - 1, j - 1)
return max(helper(i - 1, j), helper(i, j - 1))
return helper(len(seq1), len(seq2))
# ============================================================
# 2. TOP-DOWN DP (memoized recursion)
# Plain English: same idea as brute force, but REMEMBERS answers
# it has already worked out, so nothing is recomputed twice.
# ============================================================
def memoized_lcs(seq1, seq2):
"""Recursive LCS length with a cache. Time complexity: O(m*n)."""
memo = {}
def helper(i, j):
if i == 0 or j == 0:
return 0
if (i, j) in memo:
return memo[(i, j)]
if seq1[i - 1] == seq2[j - 1]:
result = 1 + helper(i - 1, j - 1)
else:
result = max(helper(i - 1, j), helper(i, j - 1))
memo[(i, j)] = result
return result
return helper(len(seq1), len(seq2))
# ============================================================
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def reconstruct_lcs(dp, seq1, seq2):
"""Walks BACKWARDS through the filled table (from the bottom-right
corner to the top-left) to recover the actual subsequence, not
just its length."""
i, j = len(seq1), len(seq2)
result = []
while i > 0 and j > 0:
if seq1[i - 1] == seq2[j - 1]:
result.append(seq1[i - 1]) # part of the LCS -> move diagonally
i -= 1
j -= 1
elif dp[i - 1][j] >= dp[i][j - 1]:
i -= 1 # move up
else:
j -= 1 # move left
result.reverse()
return "".join(result) if isinstance(seq1, str) else result
def lcs(seq1, seq2):
"""Main entry point.
Returns: (length, subsequence, dp_table)"""
dp = build_lcs_table(seq1, seq2)
length = dp[len(seq1)][len(seq2)]
subsequence = reconstruct_lcs(dp, seq1, seq2)
return length, subsequence, dp
# ============================================================
# 4. HELPER: pretty-print the DP table (great for understanding it)
# ============================================================
def print_dp_table(dp, seq1, seq2):
header = " " + " ".join(f"{c:>2}" for c in seq2)
print(header)
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117
118 # ============================================================
119 # 4. HELPER: pretty-print the DP table (great for understanding it)
120 # ============================================================
121 def print_dp_table(dp, seq1, seq2):
122 header = " " + " ".join(f"{c:>2}" for c in seq2)
123 print(header)
124 for i, row in enumerate(dp):
125 label = seq1[i - 1] if i > 0 else " "
126 print(f" {label:>2} | " + " ".join(f"{v:>2}" for v in row))
127
128
129 # ============================================================
130 # 5. DEMO / DRIVER CODE
131 # ============================================================
132 if __name__ == "__main__":
133
134 # ---- Example 1: classic textbook strings ------------------------
135 seq1 = "ABCBDAB"
136 seq2 = "BDCABA"
137
138 print(f"Sequence 1: {seq1}")
139 print(f"Sequence 2: {seq2}\n")
140
141 length, subsequence, dp = lcs(seq1, seq2)
142
143 print("DP Table (dp[i][j] = LCS length of seq1[:i] and seq2[:j]):")
144 print_dp_table(dp, seq1, seq2)
145
146 print(f"\nLCS length : {length}")
147 print(f"LCS subsequence : {subsequence}")
148
149 # ---- cross-check the three approaches agree ----------------------
150 print(f"\n[check] memoized_lcs length : {memoized_lcs(seq1, seq2)}")
151 print(f"[check] brute_force_lcs length (small input 'ABC' vs 'AC') : "
152 f"{brute_force_lcs('ABC', 'AC')}")
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138 print(f"Sequence 1: {seq1}")
139 print(f"Sequence 2: {seq2}\n")
140
141 length, subsequence, dp = lcs(seq1, seq2)
142
143 print("DP Table (dp[i][j] = LCS length of seq1[:i] and seq2[:j]):")
144 print_dp_table(dp, seq1, seq2)
145
146 print(f"\nLCS length : {length}")
147 print(f"LCS subsequence : {subsequence}")
148
149 # ---- cross-check the three approaches agree ----------------------
150 print(f"\n[check] memoized_lcs length : {memoized_lcs(seq1, seq2)}")
151 print(f"[check] brute_force_lcs length (small input 'ABC' vs 'AC') : "
152 f"{brute_force_lcs('ABC', 'AC')}")
153
154 # ---- Example 2: LCS also works on lists, not just strings ---------
155 list1 = ["walk", "to", "the", "store", "and", "buy", "milk"]
156 list2 = ["drive", "to", "store", "buy", "some", "milk"]
157 length2, subsequence2, _ = lcs(list1, list2)
158
159 print("\nWord-list example:")
160 print(f"List 1 : {list1}")
161 print(f"List 2 : {list2}")
162 print(f"LCS length : {length2}")
163 print(f"LCS words : {subsequence2}")
164
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Assignment 3.py
Users > HP > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AC8C44A748FF07C37CCCAA67D34A64146085A2BE > transfers > 2026-36 > lcs.py
Problems 33
Output
Debug Console
Terminal
Ports
Code
[check] brute_force_lcs length (small input 'ABC' vs 'AC') : 2
Word-list example:
List 1 : ['walk', 'to', 'the', 'store', 'and', 'buy', 'milk']
List 2 : ['drive', 'to', 'store', 'buy', 'some', 'milk']
LCS length : 4
LCS words : ['to', 'store', 'buy', 'milk']
PS C:\Users\HP> python -u "c:\Users\HP\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\AC8C44A748FF07C37CCCAA67D34A64146085A2BE\transfers\2026-36\lcs.py"
Sequence 1: ABCBDAB
Sequence 2: BDCABA
DP Table (dp[i][j] = LCS length of seq1[:i] and seq2[:j]):
B D C A B A
| 0 0 0 0 0 0 0
A | 0 0 0 0 1 1 1
B | 0 1 1 1 1 2 2
C | 0 1 1 2 2 2 2
B | 0 1 1 2 2 3 3
D | 0 1 2 2 2 3 3
A | 0 1 2 2 3 3 4
B | 0 1 2 2 3 4 4
LCS length : 4
LCS subsequence : BCBA
[check] memoized_lcs length : 4
[check] brute_force_lcs length (small input 'ABC' vs 'AC') : 2
Word-list example:
List 1 : ['walk', 'to', 'the', 'store', 'and', 'buy', 'milk']
List 2 : ['drive', 'to', 'store', 'buy', 'some', 'milk']
LCS length : 4
LCS words : ['to', 'store', 'buy', 'milk']
PS C:\Users\HP>
Launchpad
Ln 17, Col 70
Spaces: 4
UTF-8
LF
Python
Sign In
Python 3.14.3
Go Live
27°C
Partly sunny
Search
ENG
IN
18:04
06-09-2026